Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "electric field" and "charge" to explain your answer. Sentence starters are optional.

electric field: force per positive test charge at a location

charge: a property responsible for electric interactions

vector: a quantity with size and direction

01 Warm-up: bring back an earlier idea

A ray enters a medium with speed **1/3** of its original speed at an angle. **Sketch the bend** and state whether frequency changes.

Word help: frequency = number of cycles each second

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=**4 N/C** right, a **+2 C** test charge feels **8 N** right. A-**2 C** charge feels **8 N** left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

Sentence starter: In my work, I used … to show … . I labeled it … .

Field / first contribution

Force / second contribution

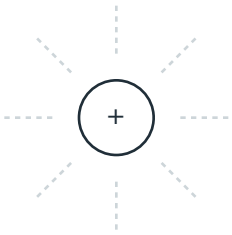
Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

An electric-field simulation uses a positive **2 C** test charge that feels **8 N** right.
1. **Find the field magnitude** and direction.
2. **Draw separate field** and force arrows and **label their different units**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

- Field / first contribution
- Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

05 Explain your reasoning

If you remove the test charge, does the source charge's field disappear? **Explain the role of a** test charge.

Sentence starter: I think … . I can point to …, which shows …, so … .

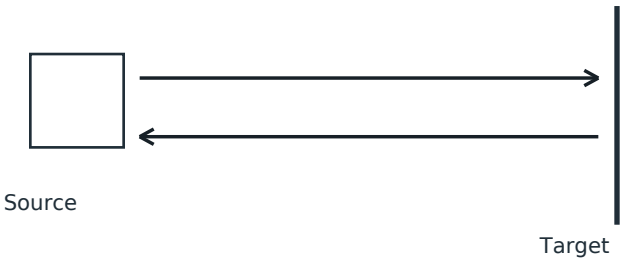
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An echo travels **120 m** on its full out-and-back path. **Find the wall's distance**.
Label the outgoing and returning paths.

Word help: echo = a sound received after reflection

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A field points down. Which way is the force on a positive charge? On a negative charge?

Sentence starter: I notice … . This supports … because … .

- Field / first contribution
- Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. **Choose two different charge values** giving forces with equal magnitudes and opposite directions in the same field.

Sentence starter: Both … . They differ in … because … .

- Field / first contribution
- Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3